LESSON 5.9

Area of Regular Polygons

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a regular polygon by decomposing it into triangles.

WHY THIS MATTERS Your architecture firm is designing a hexagonal skylight for the lobby of a new museum. The glass fabricator needs to know the total area so they can cut the correct amount of glass. Each side of the hexagon is 6 feet, and the height of each triangle formed from the center is 5.2 feet. The height is rounded to the nearest tenth of a foot, so the area is approximate.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Regular Polygon *Spanish: Polígono regular*	A closed flat shape made of straight sides, where every side is the same length and every angle is the same size.	A stop sign is a regular octagon — all 8 sides are the same length and all 8 angles are equal
Decompose *Spanish: Descomponer*	To break a shape into smaller, simpler shapes.	Draw lines from the center of a hexagon to each corner → you get 6 equal triangles you can find the area of
Triangle *Spanish: Triángulo*	A shape with three sides.	Each triangle inside a decomposed hexagon has a base = one side of the hexagon and height = distance from center to side
Composite *Spanish: Compuesto*	Made by putting two or more simple shapes together.	6 triangles, each with area 15 sq ft, combine to form a hexagon with total area = 6 × 15 = 90 sq ft
Formula *Spanish: Fórmula*	A math rule written with symbols.	For a regular hexagon: Total Area = 6 × (½ × b × h), where b = side length and h = height of each triangle

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

My hexagon has 6 equal sides. Each side is 6 feet, and each center triangle has a height of 5.2 feet. Use the rounded height to estimate the area.

1. First I find one triangle: A = ½ × base × height = ½ × 6 × 5.2.
2. I multiply: 6 × 5.2 = 31.2, then ½ × 31.2 = 15.6 square feet.
3. A hexagon has 6 sides, so it has 6 equal triangles.
4. I multiply: 6 × 15.6 = 93.6. So the total area is 93.6 square feet.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller regular pentagon: each of its 5 triangles has base 4 cm and height 3 cm. Use the rounded height to estimate the area.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The total area is _____ because one triangle is _____ and there are _____ triangles.

Word bank Regular Polygon | Decompose | Triangle | Composite | Formula | triangle | half | base | height | multiply | total | number of sides

WATCH OUT!

A common mistake

A common mistake in Area of Regular Polygons is stopping after finding the area of ONE triangle and reporting that as the total area

LESSON 5.9

Area of Regular Polygons

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Each regular polygon below is cut into congruent triangles by drawing a line from the center to every corner. Sort by how many triangles it makes.

Groups: **More than 5 triangles** **5 or fewer triangles**

_____ Regular hexagon

_____ Regular octagon

_____ Equilateral triangle

_____ Square

_____ Regular pentagon

HINT Picture it: one spoke from the center to each corner. Each triangle sits on one side.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A regular hexagon is divided into 6 equal triangles from the center. Each triangle has a base of 6 ft and a height of 5.2 ft. What is the area of one triangle? Use the given heights; nonsquare heights are rounded, so their areas are estimates.

(A) 11.2 sq ft
(B) 15.6 sq ft
(C) 15.6 ft
(D) 31.2 sq ft

HINT Focus on just ONE of the 6 triangles: its base is 6 ft and its height is 5.2 ft.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A regular hexagonal skylight has 6 central triangles, each with base 6 ft and rounded height 5.2 ft. Using those measurements, what is the approximate total area?

(A) 15.6 sq ft
(B) 62.4 sq ft
(C) 93.6 sq ft
(D) 187.2 sq ft

HINT Find one central triangle's area: ½ × 6 × 5.2 = 15.6 sq ft using the rounded height.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

A regular hexagon is split into 6 triangles, each with base 4 cm and height 3.5 cm. What is the total area? Use the given heights; nonsquare heights are rounded, so their areas are estimates.

(A) 14 sq cm
(B) 21 sq cm
(C) 42 sq cm
(D) 84 sq cm

HINT Two stages hide here: first one triangle (base 4 cm, height 3.5 cm), then all 6 of them.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

How many triangles can you make from the center of a regular pentagon?

(A) 3
(B) 5
(C) 6
(D) 10

HINT 'Penta' is a clue about how many sides a pentagon has.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake (use rounded heights)

1. **Decompose the octagon** A regular octagon = 8 congruent triangles
2. **Identify one triangle** base = 3 ft, height = 4 ft
3. **Find the area of one triangle** A = ½ × 3 × 4 = 6 sq ft
4. **Find total area** Total area = 5 × 6 = 30 sq ft

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Look at Step 4. How many sides does an octagon actually have?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

The hexagon and the octagon both have triangles, but the totals are different. What two things decide the total area of a regular polygon?

SENTENCE STARTER The total area is ___ because one triangle is ___ and there are ___ triangles.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.9

Area of Regular Polygons

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a regular polygon by decomposing it into triangles.

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table: find one central triangle's area and the polygon's total area. Use the given rounded heights for estimates; the square dimensions are exact.

Polygon	Number of Triangles	Triangle Base	Triangle Height	Area of One Triangle	Total Area
Hexagon	6	6 ft	5.2 ft		
Pentagon	5	8 ft	5.5 ft		
Square	4	5 ft	2.5 ft		
Octagon	8	4 ft	4.8 ft		

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each polygon to its total area. Use the given heights; nonsquare heights are rounded, so their areas are estimates.

_____ Hexagon: 6 triangles, each ½×6×5.2

_____ Pentagon: 5 triangles, each ½×8×5.5

_____ Octagon: 8 triangles, each ½×4×4.8

_____ Square: 4 triangles, each ½×5×2.5

_____ Decagon: 10 triangles, each about ½×3×4.6

_____ Equilateral triangle: 3 central triangles, each about ½×4×1.2

(A) 93.6 sq ft

(B) 110 sq ft

(C) 76.8 sq ft

(D) 25 sq ft

(E) 69 sq ft

(F) 7.2 sq ft

3 **Write the name of the correct group on each line.**

Find the TOTAL area of each regular polygon, then sort it against the 90 sq ft cutoff. Use the given heights; nonsquare heights are rounded, so their areas are estimates.

Groups: **Total area ≥ 90 sq ft** **Total area < 90 sq ft**

_____ Pentagon: 5 triangles, base 8 ft, height 5.5 ft

_____ Hexagon: 6 triangles, base 6 ft, height 5.2 ft

_____ Octagon: 8 triangles, base 4 ft, height 4.8 ft

_____ Square: 4 triangles, base 5 ft, height 2.5 ft

4 **Mark and label each value on the number line.**

Find the central-triangle height: square area 100 sq ft, 4 triangles, base 10 ft



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake (use rounded heights)

1. **Decompose the pentagon** A regular pentagon = 5 congruent triangles
2. **Identify one triangle** base = 6 ft, height = 4 ft
3. **Find the area of one triangle** A = 6 × 4 = 24 sq ft
4. **Find total area** Total area = 5 × 24 = 120 sq ft

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

The hexagon and the octagon both have triangles, but the totals are different. What two things decide the total area of a regular polygon?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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Challenge · Extension

MASTERY CHECK
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Use total area = number of triangles × ½ × base × height to find each missing value. Nonsquare measurements are rounded; use the displayed values.

Regular Polygon	Number of Triangles	Triangle Base	Triangle Height	Total Area
Hexagon	6	9 ft		216 sq ft
Square	4		6 ft	144 sq ft
Octagon		4 ft	5 ft	80 sq ft
Decagon	10	3 ft		69 sq ft

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Area Error (use rounded heights)

1. **Decompose the hexagon** A regular hexagon = 6 triangles
2. **Find area of one triangle** b = 5 m, h = 4.3 m
3. **Calculate one triangle** A = ½ × 5 × 4.3 = 10.75 sq m
4. **Find total area** Total = 10.75 + 6 = 16.75 sq m

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A regular hexagon and a regular pentagon are each divided into congruent central triangles. In both polygons, each central triangle has area 15 sq ft. Which polygon has greater total area? Explain why the triangle count matters. The two polygons have different side lengths.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

The hexagon and the octagon both have triangles, but the totals are different. What two things decide the total area of a regular polygon?

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